HEALTHCONNECT CLINIC EXPERIENCE LAB

Week 5 - Data Analytics Track

Data Analytics to Data Science Collaboration Report

Prepared by: Victoria Ateengbe Ikhazuangbe

1. Purpose and Context

The Week 5 collaboration connects the Data Analytics track's descriptive analysis with the Data Science track's planned predictive modelling. The purpose is to transfer evidence from the dashboard, clarify field definitions and data-quality caveats, and identify candidate variables that Data Science can validate during feature engineering and baseline modelling.

2. Track Interacted With

Data Science Track

3. Information / Output Exchanged

Analytics finding	Evidence	Potential modelling use	Caveat
Previous no-show history	55.4% current no-show vs 43.5% for clean history	Candidate behavioural-history feature	Association is not proof of causation.
Booking lead time	24.8% for 0-3 days vs 60.5% for 30+ days	Candidate booking-behaviour feature / engineered band	Long lead time may represent other unobserved factors.
Distance	46.40% for <5 km vs 54.11% for 15+ km	Candidate access feature	90 distance values are missing.
Reminder channel	SMS 45.75%; Email 48.41%; WhatsApp 49.77%; None 51.39%	Candidate reminder/intervention feature	Reminder assignment may not be random.
Appointment type	Follow-up Diagnostic 51.23%; Specialist 49.75%; General 47.44%	Candidate service-type feature	Model should validate predictive value.
Waiting time	Populated for most no-shows/cancellations; averages near 24 minutes across outcomes	Exclude or tightly scope	Likely simulated/pre-assigned; could mislead model if treated as ordinary feature.

4. Why the Collaboration Was Relevant

- The Analytics findings provide an evidence-based starting point for Data Science feature engineering.
- Shared KPI and field definitions reduce the risk of inconsistent target or predictor interpretation across tracks.
- The waiting-time anomaly is important because a technically complete field may still be unsuitable for modelling.
- The handoff creates continuity from descriptive business analysis to predictive modelling, as intended in the HealthConnect project.

5. What Changed or Improved

The collaboration creates a clearer, traceable handoff from descriptive analysis to predictive work. Rather than treating all available fields as equally useful, Data Science can prioritise validation of the strongest observed patterns and document why each candidate feature is included or excluded. Analytics, in turn, provides the definitions and caveats required to make downstream modelling consistent with the business interpretation of the data.

The collaboration also improves interpretation discipline: a large segment difference is treated as a candidate predictive signal, not automatically as a causal driver.

6. Recommended Data Science Handoff

Handoff item	Analytics contribution	Data Science action	Expected output
Target definition	Consistent use of appointment_outcome and clear treatment of Cancelled vs No-Show.	Confirm modelling target and exclusions.	Documented target definition.
Prior no-show	Highlights 55.4% vs 43.5% difference.	Test predictive contribution and transformations.	Feature validation evidence.
Lead time	Highlights strong 24.8%-60.5% gradient.	Test linear/nonlinear relationship and engineered bands.	Feature usefulness evidence.
Distance	Highlights higher rate at 15+ km.	Assess predictive value and missing-value strategy.	Validated access feature.
Reminder channel	Provides channel-level outcome differences.	Test feature/interaction value cautiously.	Model evidence on reminder variables.
Waiting time	Provides data-quality warning.	Exclude or justify separately.	Cleaner feature set.

7. Cross-Track Dependency Map

Dependency	Analytics contribution	Data Science need	Project value

Dependency	Analytics contribution	Data Science need	Project value
Shared data definitions	Data-quality assessment and KPI definitions	Consistent modelling inputs	Prevents conflicting outputs
Candidate features	EDA patterns and segment findings	Feature engineering and validation	Improves evidence-based modelling
Known data anomaly	Waiting-time warning	Avoid misleading predictor use	Protects model quality
Model feedback	Later feature importance / performance	Return modelling results to Analytics	Supports deeper business interpretation

8. Collaboration Risks and Controls

Risk	Potential effect	Control
Causal overinterpretation	Model/recommendations may treat associations as causes.	Use association language and validate through modelling/operational testing.
Feature leakage or misleading variables	Inflated or unstable model performance.	Review business meaning; exclude questionable waiting time.
Definition drift	Analytics and Data Science may report different outcomes.	Use Data Dictionary and shared target definitions.
Different missing-value treatment	Different models produce inconsistent results.	Document and align treatment of distance/reminder fields.

9. Next Collaboration Step

After the Data Science baseline model is developed, the two tracks should compare model performance and validated feature importance with the descriptive patterns in the Power BI dashboard. Where the two sources of evidence agree, confidence in prioritisation increases. Where they differ, the team should revisit segmentation, feature engineering or data quality rather than assuming the dashboard or model is automatically correct.

10. Collaboration Outcome Statement

The Data Analytics track contributed evidence on the strongest observed no-show patterns, KPI definitions and critical data caveats. This collaboration strengthens project continuity by connecting business-facing descriptive analysis to predictive modelling and by establishing a shared, documented basis for feature selection and downstream interpretation.

11. Source Documents

- AnalystLab Africa - Week 5 Experience Lab Assignment.
- HealthConnect Clinic Experience Lab - Week 4 Initial Analysis Document (Data Analytics Track).
- HealthConnect Clinic No-Show Analysis Dashboard - Power BI export.